

CHAPTER 8: MORE NUMBER THEORY

TRUE OR FALSE

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|---|---|--|
| T | F | 1. Prime numbers play a very small role in cryptography. |
| T | F | 2. One of the useful features of the Chinese remainder theorem is that it provides a way to manipulate potentially very large numbers mod M in terms of tuples of smaller numbers. |
| T | F | 3. An important requirement in a number of cryptographic algorithms is the ability to choose a large prime number. |
| T | F | 4. All integers have primitive roots. |
| T | F | 5. An area of ongoing research is the development of efficient algorithms for determining if a randomly chosen large integer is a prime number. |
| T | F | 6. The first assertion of the CRT, concerning arithmetic operations, follows from the rules for modular arithmetic. |
| T | F | 7. Discrete logarithms are not fundamental to public-key algorithms. |
| T | F | 8. The number 37 is prime so therefore all of the positive integers from 1 to 36 are relatively prime to 37. |
| T | F | 9. Discrete logarithms are analogous to ordinary logarithms but are defined using modular arithmetic. |
| T | F | 10. The Chinese Remainder Theorem is believed to have been discovered by the Chinese mathematician Agrawal in 100 A.D. |
| T | F | 11. The primitive roots for the prime number 19 are 2, 3, 10, 13, 14 and 15. |
| T | F | 12. With ordinary positive real numbers the logarithm function is the inverse of exponentiation. |
| T | F | 13. A prime number can have a remainder when divided by positive or negative values of itself. |
| T | F | 14. The Miller-Rabin test can determine if a number is not prime but |

cannot determine if a number is prime.

- T F 15. The logarithm of a number is defined to be the power to which some positive base (except 1) must be raised in order to equal the number.

MULTIPLE CHOICE

1. A _____ number can only be divided by +/- values of itself and 1 and cannot have a remainder.

A. prime	B. composite
C. indexed	D. positive

2. An important quantity in number theory referred to as _____, is defined as the number of positive integers less than n and relatively prime to n .

A. CRT	B. Miller-Rabin
C. Euler's totient function	D. Fermat's theorem

3. Miller's test will return _____ if it fails to detect that n is not prime.

A. rejected	B. inconclusive
C. composite	D. discrete

4. Prime numbers play a _____ role in number theory.

A. minor	B. nonessential
C. critical	D. abbreviated

5. If p is prime and a is a positive integer, then $a^p = a \pmod{p}$ is an alternative form of _____ theorem.

A. Rijndael's	B. Vignere's
C. Euler's	D. Fermat's

6. Two numbers are relatively prime if they have _____ prime factors in common.
- A. some B. zero
C. multiple D. all
7. The _____ algorithm is typically used to test a large number for primality.
- A. Rijndael B. Fermat
C. Miller-Rabin D. Euler
8. The procedure TEST takes a candidate integer n as input and returns the result _____ if n is definitely not a prime.
- A. discrete B. composite
C. inconclusive D. primitive
9. Two numbers are relatively prime if they have _____ prime factors in common.
- A. zero B. two
C. several D. one
10. Discrete logarithms are fundamental to the _____ .
- A. Euler algorithm B. digital signature algorithm
C. Miller-Rabin algorithm D. Rijndael algorithm
11. The procedure TEST takes a candidate integer n as input and returns the result _____ if n may or may not be a prime.
- A. discrete B. composite
C. inconclusive D. primitive

12. If a number is the highest possible exponent to which a number can belong, it is referred to as a _____ of n .
- A. primitive root B. composite
C. discrete logarithm D. bijection
13. For any integer b and a primitive root a of prime number p we can find a unique exponent i . This exponent i is referred to as the _____.
- A. order B. discrete logarithm
C. bijection D. primitive root
14. Discrete logarithms are fundamental to a number of public-key algorithms including _____ key exchange and the DSA.
- A. Diffie-Hellman B. Rijndael-Fadiman
C. Fermat-Euler D. Miller-Rabin
15. A one-to-one correspondence is called _____.
- A. a bijection B. an inclusive
C. a composite D. an index

SHORT ANSWER

1. A _____ number is an integer that can only be divided by positive and negative values of itself and 1 without having a remainder.
2. Two theorems that play important roles in public-key cryptography are Fermat's theorem and _____ theorem.
3. Discrete logarithms are analogous to ordinary logarithms but are defined using _____ arithmetic.
4. _____ theorem states the following: If p is prime and a is a positive integer not divisible by p , then $a^{p-1} = 1 \pmod{p}$.

5. Two numbers are _____ if their greatest common divisor is 1.
6. The number of positive integers less than n and relatively prime to n is referred to as _____ function.
7. The _____ theorem states that it is possible to reconstruct integers in a certain range from their residues modulo a set of pairwise relatively prime moduli.
8. The mapping of the CRT equation is a one-to-one correspondence called a _____ between Z_m and the Cartesian product $Z_{m_1} \times Z_{m_2} \times \dots \times Z_{m_k}$.
9. Discrete logarithms are fundamental to the digital signature algorithm and the _____ algorithm.
10. The _____ of a number is defined to be the power to which some positive base (except 1) must be raised in order to equal the number.
11. To determine whether an odd integer n is prime with a reasonable degree of confidence repeatedly invoke TEST (n) using randomly chosen values for a . If, at any point, TEST returns _____ then n is determined to be nonprime.
12. Two numbers are relatively prime if their greatest common divisor is _____.
13. An integer $p > 1$ is a _____ number if and only if its only divisors are ± 1 and $\pm p$.
14. The _____ of integers a and b , expressed $(\gcd a, b)$, is an integer c that divides both a and b without remainder and that any divisor of a and b is a divisor of c .
15. Although it does not appear to be as efficient as the Miller-Rabin algorithm, in 2002 a relatively simple deterministic algorithm that efficiently determines whether a given large number is a prime was developed. This algorithm is known as the _____ algorithm.